

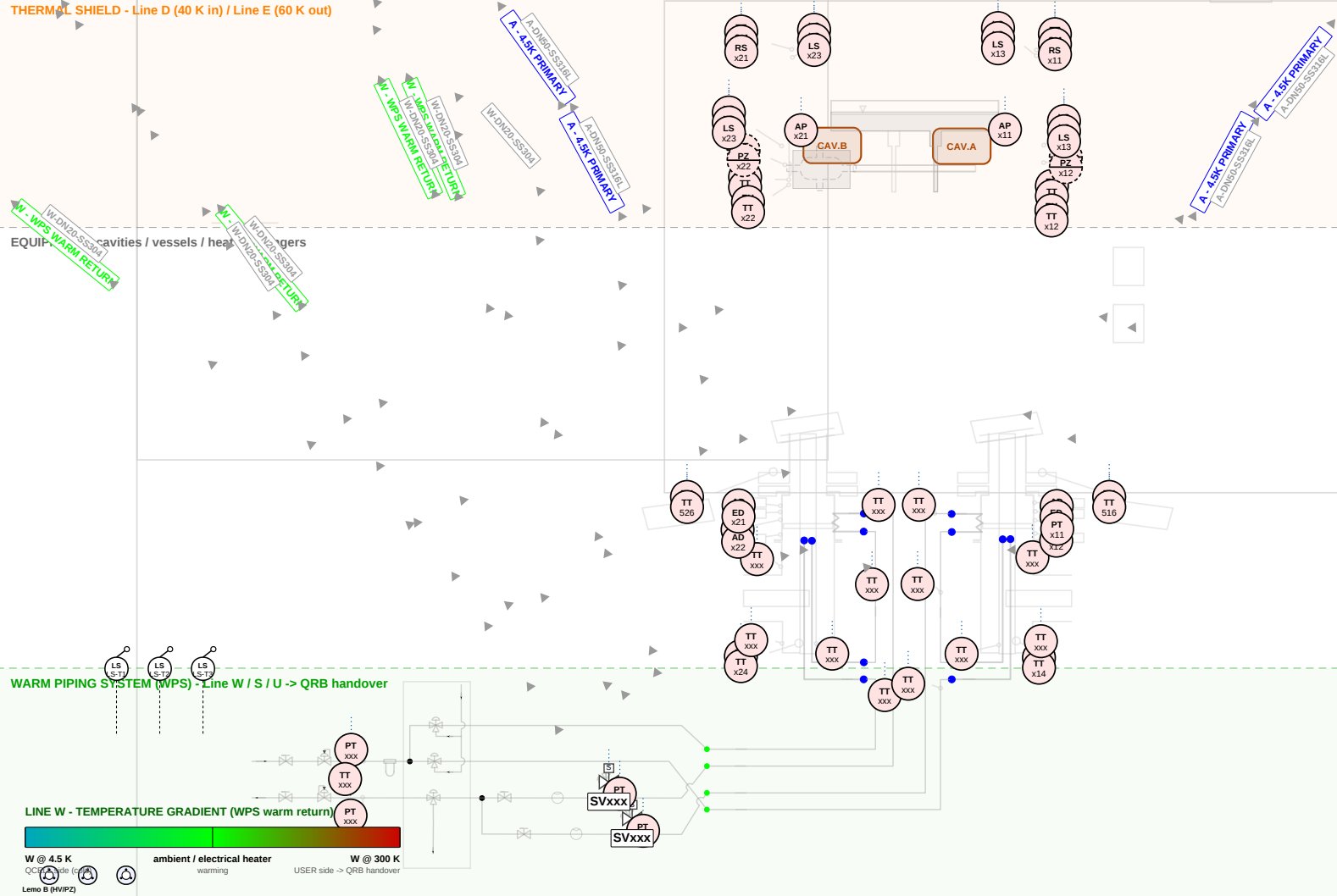
THERMAL SHIELD - Line D (40 K in) / Line E (60 K out)

EQUIPMENT: cavities / vessels / heat exchangers

WARM PIPING SYSTEM (WPS) - Line W / S / U -> QRB handover

LINE W - TEMPERATURE GRADIENT (WPS warm return)

W @ 4.5 K ambient / electrical heater W @ 300 K
 QCR side (cable) warming USER side -> QRB handover
 Lemo B (HV/PIZ)



INTERACTIVE LEGEND

toggle: layer "16_Legend_INTERACTIVE"

LINE SPECIFICATION TABLE

	LINE	TEMP	PRESS	DN	MOC
COLD HEADER	A	4.5 K	3 bar	DN50	SS316L
	A'	4.5 K	3 bar	DN25	SS316L
	B	2 K	27 mbar	DN40	SS316L
	B'	2 K	27 mbar	DN25	SS316L
THERMAL SHIELD	D	40 K	14 bar	DN32	CU
	D'	40 K	14 bar	DN20	CU
	E	60 K	13 bar	DN32	CU
	E'	60 K	13 bar	DN20	CU
WARM (WPS)	W	4.5 K-300	6 bar	DN20	SS304
	S	2-292 K	1.05 bar	DN15	SS304
	U	292 K	14 bar	DN15	SS304
REFERENCE					
	OS	-	-	-	-

mono: colour -> line weight + dash + on-line NAME LABEL

SIGNAL TYPES (0.25 mm)

Pneumatic (dash + //)

Electric (dotted)

Hydraulic (dash-dot)

INSTRUMENTS / SCOPE

TT

Field instrument (white box tag)

PNE

Panel / safety (line)

SCOPE

TP scope diamond (edge = TP layer)

Cat: B=Building, C=Chil, E=Electrical, G=Compressed gasses

HH=HVAC, L=Liquid waste, S=Solid waste, W=Water

CONSULTANT Mott MacDonald Bristol, UK		CLIENT	REVISIONS			APPROVALS		MINERVA CryoCell - SCK CEN RFCELL (seen by ACR) - Instrumentation & C DWG No. = NA.P501_PFB713 MMD Proj. 411066 Scale NTS Suitability S2 - FOR ACCEPTANCE Security RESTRICTED Size / Std A3 / ISO19628 / ISA-9.1 Sheet Sheet2 of 2 - Instrumentation Variant CONTROL-CENTRIC	
		SCK CEN	Rev	Date	Description	App'd	Designed		QSYS
		Boerentang 200, 2400 Mol, BE	C1	2026-06	v3 refinement - layers/legibil	ACR	Drawn		ACR
		MYRRHA / MINERVA Phase 1	B1	2026-05	v2 split sheets	ACR	Checked		ACR
			A1	2026-04	First issue	ACR	Approved		PM